Geometric Basin Connectivity in the Large-b Regime: Exact Spherical Analysis

Abstract

We study geometric connectivity of pattern basins in the LSR associative memory when the nonlinearity parameter scales as $b = \beta N$ with $\beta > 0$ fixed. The analysis proceeds in two stages: first purely geometric (do the spherical caps overlap at all?), then energetic (can the state traverse the corridor?). For $b = \beta N$, we show that energy barriers between geometrically overlapping basins are $O(1/\beta)$, so geometry is the sole bottleneck. We then compute the expected number of geometric neighbours λ using the *exact* spherical overlap distribution and compare with the Gaussian (Mill's ratio) approximation. The two predictions differ by a super-exponential factor: the Gaussian gives a percolation threshold $\alpha_c^{\text{Gauss}} \to 1/2$, while the exact sphere gives $\alpha_c^{\text{sphere}} \approx \frac{1}{2}\ln(\beta N/8)$, which diverges with N. We conclude that for $b = \beta N$ and large N, every basin is geometrically isolated at any fixed load α. The phase transition at $\alpha = 1/2$ in the exponential associative memory is driven by signal-to-noise competition in the softmax energy, not by basin connectivity.

1 The model

The Log-Sum-ReLU (LSR) associative memory stores M patterns $\boldsymbol{\xi}^1, \ldots, \boldsymbol{\xi}^M$ drawn uniformly on the sphere $S^{N-1}(\sqrt{N})$ (i.e. $|\boldsymbol{\xi}^\mu|^2 = N$). The energy of a state $\mathbf{S} \in S^{N-1}(\sqrt{N})$ is

$$E(\mathbf{S}) = -\frac{N}{b}\sum_{\mu=1}^{M} \ln\big[\,1 - b + b\,\varphi_\mu(\mathbf{S})\,\big]_+\,, \tag{1}$$

where $\varphi_\mu(\mathbf{S}) = (\boldsymbol{\xi}^\mu \cdot \mathbf{S})/N$ is the overlap between the state and pattern μ, $[\cdot]_+ = \max(0, \cdot)$, and $b > 0$ is the nonlinearity parameter. The number of patterns scales exponentially with dimension: $M = e^{N\alpha}$, where $\alpha > 0$ is the load.

Throughout this note we work in the regime

$$b = \beta N, \qquad \beta > 0 \text{ fixed}, \qquad \beta N \gg 1. \tag{2}$$

1.1 Hard-wall threshold and basins

The argument of the logarithm in (1) is $1 - b + b\,\varphi_\mu = b(\varphi_\mu - \varphi_c)$, where

$$\varphi_c \;=\; \frac{b-1}{b} \;=\; 1 - \frac{1}{b} \;=\; 1 - \frac{1}{\beta N}\,. \tag{3}$$

Pattern μ contributes to the energy only when $\varphi_\mu(\mathbf{S}) > \varphi_c$; otherwise $[\cdots]_+ = 0$. If *no* pattern is active, the sum is empty and $E = +\infty$ (the state is energetically forbidden).

The **basin** of pattern μ is the spherical cap

$$X_\mu = \big\{\, \mathbf{S} \in S^{N-1}(\sqrt{N}) \;:\; \varphi_\mu(\mathbf{S}) > \varphi_c \,\big\}. \tag{4}$$

The boundary $\varphi_\mu = \varphi_c$ is an infinite energy barrier ($\ln 0^+ = -\infty$ in the energy, so $E \to +\infty$). Within a single basin, the state is permanently confined.

1.2 Angular radius of a basin

The condition $\varphi_\mu > \varphi_c$ is equivalent to $\cos\theta < \varphi_c$, where θ is the angle between $\mathbf{S}$ and $\boldsymbol{\xi}^\mu$. The cap has angular radius

$$\theta_c = \arccos\varphi_c = \arccos\Big(1 - \frac{1}{\beta N}\Big) \approx \sqrt{\frac{2}{\beta N}} \qquad (\beta N \gg 1), \tag{5}$$

using $\arccos(1-x) \approx \sqrt{2x}$ for $x \ll 1$. So each basin is a tiny cap whose angular radius shrinks as $1/\sqrt{\beta N}$.

2 Geometric overlap of two basins

2.1 Criterion for cap intersection

Two spherical caps of equal angular radius θ_c overlap if and only if the angle $\theta_{\mu\nu}$ between their centres satisfies $\theta_{\mu\nu} < 2\theta_c$. In terms of the mutual overlap $q = (\boldsymbol{\xi}^\mu \cdot \boldsymbol{\xi}^\nu)/N = \cos\theta_{\mu\nu}$, the caps overlap when

$$q > q_{\text{geom}} \;\equiv\; \cos(2\theta_c) = 2\cos^2\theta_c - 1 = 2\varphi_c^2 - 1. \tag{6}$$

Substituting $\varphi_c = 1 - 1/(\beta N)$:

$$q_{\text{geom}} = 2\Big(1 - \frac{1}{\beta N}\Big)^2 - 1 = 1 - \frac{4}{\beta N} + \frac{2}{\beta^2 N^2}\,. \tag{7}$$

This is extremely close to 1: the two patterns must be nearly identical for their basins to touch.

If $q < q_{\text{geom}}$, the caps are disjoint and no continuous path connects them without leaving all basins (hitting $E = +\infty$). No amount of thermal energy can bridge a gap where no corridor exists.

2.2 The corridor between overlapping basins

When $q > q_{\text{geom}}$, the caps X_μ and X_ν share a non-empty intersection: the **corridor** $X_\mu \cap X_\nu$. Along any path through the corridor, at least two patterns are active, and the energy remains finite.

To compute the energy along the corridor, consider the geodesic (great-circle arc) from $\boldsymbol{\xi}^\mu$ to $\boldsymbol{\xi}^\nu$ on $S^{N-1}(\sqrt{N})$. Parametrise it by the angle $\psi \in [0, \theta_{\mu\nu}]$ from $\boldsymbol{\xi}^\mu$:

$$\mathbf{S}(\psi) = \frac{\sin(\theta_{\mu\nu} - \psi)}{\sin\theta_{\mu\nu}}\,\boldsymbol{\xi}^\mu + \frac{\sin\psi}{\sin\theta_{\mu\nu}}\,\boldsymbol{\xi}^\nu. \tag{8}$$

The overlaps along this geodesic are

$$\varphi_\mu(\psi) = \frac{\boldsymbol{\xi}^\mu \cdot \mathbf{S}(\psi)}{N} = \frac{\sin(\theta_{\mu\nu} - \psi)}{\sin\theta_{\mu\nu}} + q\,\frac{\sin\psi}{\sin\theta_{\mu\nu}}\,, \tag{9}$$

$$\varphi_\nu(\psi) = \frac{\boldsymbol{\xi}^\nu \cdot \mathbf{S}(\psi)}{N} = q\,\frac{\sin(\theta_{\mu\nu} - \psi)}{\sin\theta_{\mu\nu}} + \frac{\sin\psi}{\sin\theta_{\mu\nu}}\,. \tag{10}$$

At $\psi = 0$ (centre of μ): $\varphi_\mu = 1$, $\varphi_\nu = q$. At $\psi = \theta_{\mu\nu}$ (centre of ν): $\varphi_\mu = q$, $\varphi_\nu = 1$.

3 Energy barrier along the geodesic

3.1 Energy of a single active pattern

When only pattern μ is active (with overlap $\varphi_\mu > \varphi_c$), its contribution to the energy is

$$E_\mu = -\frac{N}{b}\,\ln\bigl[1 - b + b\,\varphi_\mu\bigr] = -\frac{N}{b}\,\ln\bigl[b(\varphi_\mu - \varphi_c)\bigr]. \tag{11}$$

At the basin centre ($\varphi_\mu = 1$):

$$E_\mu^{\text{centre}} = -\frac{N}{b}\,\ln\bigl[b(1-\varphi_c)\bigr] = -\frac{N}{b}\,\ln 1 = 0\,, \tag{12}$$

using $b(1-\varphi_c) = b\cdot(1/b) = 1$.

At the basin boundary ($\varphi_\mu \to \varphi_c^+$): $E_\mu \to -\frac{N}{b}\ln 0^+ = +\infty$.

3.2 Energy at the bottleneck

The energy is highest at the "bottleneck"—the point on the geodesic where one pattern has just activated. By symmetry of the barrier calculation, consider the point ψ^* where pattern ν just activates: $\varphi_\nu(\psi^*) = \varphi_c$. At this point, ν contributes $\ln(0^+ + 1) = 0$ to the energy, and the entire energy comes from pattern μ:

$$E(\psi^*) = -\frac{N}{b}\,\ln\bigl[b(\varphi_\mu^* - \varphi_c)\bigr], \tag{13}$$

where φ_μ^* is the overlap with μ when $\varphi_\nu = \varphi_c$.

To find φ_μ^*, we use a coordinate decomposition. Write $\boldsymbol{\xi}^\nu = q\,\boldsymbol{\xi}^\mu + \sqrt{1-q^2}\,\hat{\boldsymbol{\xi}}^\perp$, where $\hat{\boldsymbol{\xi}}^\perp$ is a unit vector (rescaled to $|\hat{\boldsymbol{\xi}}^\perp|^2 = N$) orthogonal to $\boldsymbol{\xi}^\mu$. Then at any state $\mathbf{S}$:

$$\varphi_\nu = q\,\varphi_\mu + \sqrt{1-q^2}\,\varphi_\perp, \tag{14}$$

where $\varphi_\perp = (\hat{\boldsymbol{\xi}}^\perp \cdot \mathbf{S})/N$. Setting $\varphi_\nu = \varphi_c$ and solving for φ_μ on the geodesic (where $\varphi_\mu^2 + \varphi_\perp^2 \le 1$ with appropriate normalisation), one finds the overlap with μ at the ν-activation boundary. On the geodesic in the $(\varphi_\mu, \varphi_\perp)$ plane, the state satisfies $\varphi_\mu = \cos\psi$, $\varphi_\perp = \sin\psi\cdot\sqrt{1-q^2}/\sin\theta_{\mu\nu}$ (to leading order for small angles).

The result is:

$$\boxed{\varphi_\mu^* = q\,\varphi_c + \sqrt{(1-q^2)(1-\varphi_c^2)}\,.} \tag{15}$$

This can be verified geometrically: the point where the geodesic from $\boldsymbol{\xi}^\mu$ to $\boldsymbol{\xi}^\nu$ crosses the boundary of cap X_ν (the circle $\varphi_\nu = \varphi_c$) has overlap φ_μ^* with pattern μ.

Check at $q = q_{\text{geom}}$. When the caps just touch ($q = 2\varphi_c^2 - 1$):

$$\begin{aligned}
\varphi_\mu^* &= (2\varphi_c^2 - 1)\varphi_c + \sqrt{(1-(2\varphi_c^2-1)^2)(1-\varphi_c^2)} \\
&= (2\varphi_c^2 - 1)\varphi_c + \sqrt{4\varphi_c^2(1-\varphi_c^2)\cdot(1-\varphi_c^2)} \\
&= (2\varphi_c^2 - 1)\varphi_c + 2\varphi_c(1-\varphi_c^2) \\
&= \varphi_c\bigl[2\varphi_c^2 - 1 + 2 - 2\varphi_c^2\bigr] = \varphi_c\,.
\end{aligned} \tag{16}$$

So $\varphi_\mu^* = \varphi_c$: both patterns are marginal, confirming the caps just touch.

3.3 The barrier height

The **barrier** is the energy difference between the bottleneck and the basin centre:

$$\boxed{\Delta E = E(\psi^*) - E_\mu^{\text{centre}} = -\frac{N}{b}\ln\big[b(\varphi_\mu^* - \varphi_c)\big] = \frac{N}{b}\,\ln\frac{1}{b\,(\varphi_\mu^* - \varphi_c)}\,.} \tag{17}$$

The barrier diverges as $q \to q_{\text{geom}}^+$ (where $\varphi_\mu^* \to \varphi_c$), confirming that caps which just barely touch have an infinite barrier.

3.4 Barrier scaling for $b = \beta N$

For $b = \beta N$, the prefactor is $N/b = 1/\beta$. The overlap excess is

$$\varphi_\mu^* - \varphi_c = q\,\varphi_c + \sqrt{(1-q^2)(1-\varphi_c^2)} - \varphi_c = \varphi_c(q-1) + \sqrt{(1-q^2)(1-\varphi_c^2)}\,. \tag{18}$$

For q slightly above q_{geom}, write $q = q_{\text{geom}} + \epsilon$ with $\epsilon \ll 1$. Using $1 - \varphi_c^2 \approx 2/(\beta N)$ and $1 - q_{\text{geom}} \approx 4/(\beta N)$:

$$\varphi_\mu^* - \varphi_c \;\approx\; \frac{\epsilon}{2\varphi_c} \;\approx\; \frac{\epsilon}{2}\,. \tag{19}$$

The argument of the logarithm in (18):

$$b(\varphi_\mu^* - \varphi_c) \approx \beta N \cdot \frac{\epsilon}{2} = \frac{\beta N\,\epsilon}{2}\,. \tag{20}$$

For a "typical" geometrically overlapping pair, $\epsilon = O(1/(\beta N))$ (the overlap just barely exceeds the threshold), so $b(\varphi_\mu^* - \varphi_c) = O(1)$ and

$$\boxed{\Delta E = \frac{1}{\beta}\,\ln\frac{1}{\beta N\,(\varphi_\mu^* - \varphi_c)} = O(1/\beta)\,.} \tag{21}$$

This is a finite, N-independent barrier. Any temperature $T > 0$ (even $T \ll 1$) overcomes it on an $O(1)$ timescale, via Kramers-type thermal activation $\tau \sim e^{\Delta E/T}$ with $\Delta E/T = O(1)$.

Contrast with $b = O(1)$. For $b = O(1)$, the prefactor $N/b = O(N)$, giving $\Delta E = O(N)$: an extensive barrier that grows with system size and is thermally impassable at any fixed T.

Conclusion. For $b = \beta N$, **geometry is the sole bottleneck**. If two caps overlap ($q > q_{\text{geom}}$), the barrier is $O(1/\beta)$ and thermally irrelevant. If the caps do not overlap ($q < q_{\text{geom}}$), no corridor exists and no amount of thermal energy helps. We therefore focus entirely on the geometric question: how many caps overlap?

4 Overlap distribution on the sphere

For two independent random points on $S^{N-1}(\sqrt{N})$, the normalised overlap $q = (\boldsymbol{\xi}^\mu \cdot \boldsymbol{\xi}^\nu)/N$ is the cosine of the angle between them. It has the exact marginal density

$$f(q) = \frac{(1-q^2)^{(N-3)/2}}{B\big(\frac{1}{2},\,\frac{N-1}{2}\big)}\,, \qquad q \in (-1,1), \tag{22}$$

where $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$ is the beta function.

Derivation. Fix $\boldsymbol{\xi}^\mu$ (say, along the first coordinate axis). The overlap $q = \xi_1^\nu/\sqrt{N}$ (the first component of $\boldsymbol{\xi}^\nu$, rescaled). For a uniform point on $S^{N-1}(\sqrt{N})$, the projection onto any axis has the distribution of $\sqrt{N}\cos\Theta$ where Θ is the polar angle. The surface area element at polar angle Θ on S^{N-1} is proportional to $\sin^{N-2}\Theta\, d\Theta$. Changing variables to $q = \cos\Theta$: $dq = -\sin\Theta\, d\Theta$ and $\sin^{N-2}\Theta = (1-q^2)^{(N-2)/2}$. Thus $f(q) \propto (1-q^2)^{(N-2)/2}/\sin\Theta = (1-q^2)^{(N-2)/2}/(1-q^2)^{1/2} = (1-q^2)^{(N-3)/2}$. The normalisation constant is $1/B(1/2,(N-1)/2)$ by the beta-function integral $\int_{-1}^{1}(1-q^2)^{(N-3)/2}\, dq = B(1/2,(N-1)/2)$.

Moments. $\mathbb{E}[q] = 0$ (by symmetry), $\mathbb{E}[q^2] = 1/(N-1) \approx 1/N$.

Gaussian approximation. Matching the first two moments, one approximates $q \sim \mathcal{N}(0, 1/N)$. This is accurate for $q = O(1/\sqrt{N})$ but breaks down catastrophically in the tail near $q = 1$, as we now show.

5 Tail probability: sphere vs. Gaussian

The expected number of geometric neighbours depends on the tail probability $P(q > q_{\text{geom}})$.

5.1 Exact spherical tail near $q = 1$

For q_0 close to 1, set $q = 1 - t$ with $\delta = 1 - q_0 \ll 1$:

$$1 - q^2 = (1-q)(1+q) = t(2-t) \approx 2t \qquad (t \ll 1). \tag{23}$$

Then:

$$\begin{aligned} P_{\text{sphere}}(q > q_0) &= \int_{q_0}^{1} f(q)\, dq = C_N \int_0^{\delta} (2t)^{(N-3)/2}\, dt \\ &= C_N \cdot 2^{(N-3)/2} \cdot \frac{\delta^{(N-1)/2}}{(N-1)/2}\,, \end{aligned} \tag{24}$$

where $C_N = 1/B(1/2,(N-1)/2)$.

For the normalisation constant, use Stirling's approximation on the beta function:

$$B\Big(\frac{1}{2}, \frac{N-1}{2}\Big) = \frac{\sqrt{\pi}\,\Gamma\big(\frac{N-1}{2}\big)}{\Gamma\big(\frac{N}{2}\big)} \approx \sqrt{\frac{2\pi}{N-1}} \approx \sqrt{\frac{2\pi}{N}}\,, \tag{25}$$

so $C_N \approx \sqrt{N/(2\pi)}$.

Combining:

$$P_{\text{sphere}} \approx \sqrt{\frac{N}{2\pi}} \cdot \frac{2^{(N-3)/2}\,\delta^{(N-1)/2}}{(N-1)/2} = \frac{(2\delta)^{(N-1)/2}}{(N-1)\sqrt{2\delta}} \cdot \sqrt{\frac{2N}{\pi}}\,. \tag{26}$$

The leading exponential is:

$$\boxed{\ln P_{\text{sphere}} = \frac{N-1}{2}\,\ln(2\delta) + O(\ln N)\,,} \tag{27}$$

where $\delta = 1 - q_{\text{geom}} = 4/(\beta N)$, giving

$$\ln P_{\text{sphere}} = \frac{N-1}{2}\,\ln\frac{8}{\beta N} + O(\ln N). \tag{28}$$

Since $\beta N \gg 1$, we have $\ln(8/(\beta N)) < 0$ and P_{sphere} is exponentially small—but the exponent is $(N/2)\ln(8/(\beta N))$, not $-N/2$.

5.2 Gaussian tail

Under $q \sim \mathcal{N}(0, 1/N)$, the standardised variable is $z = q\sqrt{N}$ and $z_0 = q_{\text{geom}}\sqrt{N} \approx \sqrt{N} - 4/(\beta\sqrt{N})$. By Mill's ratio ($\bar{\Phi}(z) \approx \phi(z)/z$ for $z \gg 1$):

$$P_{\text{Gauss}} = \bar{\Phi}(z_0) \approx \frac{e^{-z_0^2/2}}{z_0\sqrt{2\pi}}\,. \tag{29}$$

The exponent:

$$-\frac{z_0^2}{2} = -\frac{N\,q_{\text{geom}}^2}{2} \approx -\frac{N}{2}\Big(1 - \frac{8}{\beta N}\Big) = -\frac{N}{2} + \frac{4}{\beta}\,, \tag{30}$$

so

$$\boxed{\ln P_{\text{Gauss}} = -\frac{N}{2} + \frac{4}{\beta} + O(\ln N)\,.} \tag{31}$$

5.3 The discrepancy

Taking the ratio:

$$\begin{aligned}
\ln\frac{P_{\text{sphere}}}{P_{\text{Gauss}}} &= \frac{N}{2}\ln\frac{8}{\beta N} - \Big(-\frac{N}{2} + \frac{4}{\beta}\Big) + O(\ln N) \\
&= \frac{N}{2}\Big(\ln\frac{8}{\beta N} + 1\Big) - \frac{4}{\beta} + O(\ln N) \\
&= -\frac{N}{2}\Big(\ln\frac{\beta N}{8} - 1\Big) + O(\ln N).
\end{aligned} \tag{32}$$

$$\boxed{\frac{P_{\text{sphere}}}{P_{\text{Gauss}}} \;\approx\; \exp\Big[-\frac{N}{2}\Big(\ln\frac{\beta N}{8} - 1\Big)\Big].} \tag{33}$$

For $\beta N > 8e \approx 21.7$, the exponent is negative and the sphere gives a *smaller* tail than the Gaussian. The discrepancy grows as $(N/2)\ln(\beta N)$—this is **super-exponential** in N.

Why the Gaussian fails. The exact log-density is

$$\ln f(q) = \text{const} + \frac{N-3}{2}\ln(1 - q^2). \tag{34}$$

Expanding around $q = 0$:

$$\frac{N-3}{2}\ln(1 - q^2) = -\frac{N-3}{2}\,q^2 - \frac{N-3}{4}\,q^4 - \frac{N-3}{6}\,q^6 - \cdots \tag{35}$$

The Gaussian keeps only the q^2 term. For $q = O(1/\sqrt{N})$, the q^4 correction is $O(1)$ and can be absorbed into the normalisation. But for $q \to 1$, *every term is* $O(N)$: $q^{2k} \approx 1$ and the k-th term is $-(N-3)/(2k)$. The full series sums to the exact $\ln(1 - q^2)$, which diverges as $q \to 1$—qualitatively different from the finite Gaussian exponent $-Nq^2/2 \approx -N/2$.

6 Geometric percolation threshold

6.1 Expected number of geometric neighbours

The expected number of patterns whose basins overlap with a given pattern's basin is

$$\lambda = (M-1) \cdot P(q > q_{\text{geom}}) \approx e^{N\alpha} \cdot P. \tag{36}$$

When $\lambda > 1$, a typical pattern has at least one geometric neighbour; when $\lambda \gg 1$, a giant connected component forms in the random geometric graph on the sphere. The percolation threshold $\lambda = 1$ gives $\alpha_c = -(\ln P)/N$.

6.2 Gaussian prediction

From (32):

$$\alpha_c^{\text{Gauss}} = \frac{1}{2} - \frac{4}{\beta N} + \frac{O(\ln N)}{N} \xrightarrow{N\to\infty} \frac{1}{2}\,. \tag{37}$$

This is the standard result cited for the exponential associative memory.

6.3 Exact spherical prediction

From (29):

$$\boxed{\alpha_c^{\text{sphere}} = -\frac{1}{N} \cdot \frac{N-1}{2} \ln \frac{8}{\beta N} + \frac{O(\ln N)}{N} \approx \frac{1}{2} \ln \frac{\beta N}{8} = \frac{\ln \beta + \ln N - \ln 8}{2}\,.} \tag{38}$$

This diverges as $\frac{1}{2} \ln N$ for any fixed $\beta > 0$.

For any fixed load α, the condition $\alpha > \alpha_c^{\text{sphere}}$ is violated once

$$N > N_{\text{crit}} = \frac{8}{\beta} e^{2\alpha}. \tag{39}$$

Beyond N_{crit}, $\lambda \to 0$ super-exponentially and every basin is geometrically isolated.

6.4 Numerical comparison

N	β	βN	α_c^{Gauss}	α_c^{sphere}	$\ln(P_{\text{sphere}}/P_{\text{Gauss}})$
10	1	10	0.50	0.11	+1
20	1	20	0.50	0.46	−1
50	1	50	0.50	0.93	−22
100	1	100	0.50	1.26	−68
200	1	200	0.50	1.61	−183
50	0.2	10	0.50	0.11	+2
50	0.5	25	0.50	0.57	−4
50	2.0	100	0.50	1.26	−39

For $\beta N \lesssim 20$, the two predictions are comparable (the sphere even gives a *heavier* tail than the Gaussian for $\beta N = 10$, because the leading-order formula (28) drops polynomial prefactors that matter at small βN). For $\beta N \gtrsim 30$, the Gaussian overestimates λ by many orders of magnitude.

Example: $\alpha = 0.5$**,** $\beta = 1$**.** The Gaussian predicts $\lambda \approx 1$ (percolation threshold). The exact sphere gives $N_{\text{crit}} = 8e \approx 22$. For $N = 50$: $\lambda_{\text{sphere}} \approx e^{50 \cdot 0.5 + 25 \ln(8/50)} \approx e^{25-46} = e^{-21} \approx 10^{-9}$. The basin has zero geometric neighbours with overwhelming probability.

7 Structure of the geometric graph

7.1 When chains can form

For $\alpha > \alpha_c^{\text{sphere}} = \frac{1}{2} \ln(\beta N/8)$, the geometric random graph on S^{N-1} has $\lambda > 1$ and forms a giant connected component with chain structure:

- Each hop covers angular distance $\sim 2\theta_c \approx 2\sqrt{2/(\beta N)}$.
- To traverse $O(1)$ angular distance requires $k \sim \sqrt{\beta N}/2$ hops.
- At each step, the frontier shell (annulus of width $\sim \theta_c$ beyond the current boundary) contains $\sim \lambda$ candidate patterns.
- After k hops, the cluster contains $\sim \lambda^k$ patterns, which is sub-exponential in N for $k = O(\sqrt{N})$.

7.2 Why chains are absent at fixed α

For any fixed α and $N > N_{\text{crit}} = 8e^{2\alpha}/\beta$:

- $\lambda \to 0$ super-exponentially.
- Each basin is an isolated island with no geometric neighbours.
- The BFS from any starting state terminates at depth 0: no patterns are found beyond the initial basin.

7.3 State-space BFS at $\alpha \approx 0.5$

Starting at $s_0 \in X_{\mu_0}$ and walking to the boundary:

- $\beta N \lesssim 20$: the geometric graph is non-trivial; chains of a few hops may exist. This is the small-βN regime where the Gaussian approximation is qualitatively adequate.
- $\beta N \approx 22$: crossover. $\lambda \approx 1$.
- $\beta N \gtrsim 30$: $\lambda \ll 1$; no geometric neighbours exist. The BFS terminates immediately.

8 What drives the phase transition at $\alpha = 1/2$?

The exponential associative memory ($b \to \infty$ limit of (1)) has energy $E = -\frac{1}{\beta} \ln \sum_\mu e^{\beta N \varphi_\mu}$ and a well-established phase transition at $\alpha_c = 1/2$. Since geometric percolation requires $\alpha \gg 1/2$ for large N, this transition must be driven by a different mechanism.

8.1 Signal-to-noise analysis

At the target centre $\mathbf{S} = \boldsymbol{\xi}^1$, the energy gradient is proportional to a softmax-weighted sum of patterns:

$$\nabla E \propto \sum_\mu w_\mu \, \boldsymbol{\xi}^\mu, \qquad w_\mu = \frac{e^{\beta N \varphi_\mu}}{\sum_\nu e^{\beta N \varphi_\nu}}. \tag{40}$$

The **signal** is the target's contribution: $e^{\beta N \cdot 1} = e^{\beta N}$. The **noise** is the sum of all other contributions:

$$\text{Noise} = \sum_{\mu \neq 1} e^{\beta N \varphi_\mu}. \tag{41}$$

For random overlaps $\varphi_\mu \sim \mathcal{N}(0, 1/N)$ (in the Gaussian approximation), the moment generating function gives $\mathbb{E}[e^{\beta N \varphi}] = e^{\beta^2 N/2}$. Therefore:

$$\mathbb{E}[\text{Noise}] = (M-1)\, e^{\beta^2 N/2} \approx e^{N\alpha + \beta^2 N/2}. \tag{42}$$

The target dominates when $e^{\beta N} \gg e^{N\alpha + \beta^2 N/2}$, i.e.

$$\boxed{\alpha < \beta - \frac{\beta^2}{2} = \beta\Big(1 - \frac{\beta}{2}\Big).} \tag{43}$$

For $\beta = 1$: $\alpha_c = 1/2$. The maximum over β is at $\beta^* = 1$, giving $\alpha_c^{\max} = 1/2$.

8.2 Nature of the transition

This is an **information-theoretic** transition:

- The state does not walk to other basins. Each basin is geometrically isolated for $\beta N \gg 1$.
- Exponentially many patterns contribute to the softmax through their weights $e^{\beta N \varphi_\mu}$, even though *none* of them are "active" in the hard-wall sense (all have $\varphi_\mu < \varphi_c = 1 - 1/(\beta N)$ at the target centre).
- When $\alpha > 1/2$, the collective noise from $\sim e^{N/2}$ patterns overwhelms the target signal. The target ceases to be a stable fixed point of the dynamics.

Key distinction from $b = O(1)$. For $b = O(1)$ (the LSR regime), the hard wall at $\varphi_c = O(1)$ creates genuine infinite barriers between well-separated basins. The phase transition occurs when the state can percolate through a network of corridors. For $b = \beta N$, the hard wall at $\varphi_c = 1 - O(1/N)$ is so restrictive that basins are microscopic and isolated. The transition occurs because the softmax energy is effectively a global interaction—every pattern contributes everywhere on the sphere through its exponential weight, regardless of the hard-wall constraint.

8.3 Spherical correction to the signal-to-noise threshold

The Gaussian MGF $\mathbb{E}[e^{\beta N q}] = e^{\beta^2 N/2}$ receives corrections from the exact spherical distribution. The exact MGF is

$$\mathbb{E}_{\text{sphere}}[e^{\beta N q}] = C_N \int_{-1}^{1} e^{\beta N q} \, (1 - q^2)^{(N-3)/2} \, dq. \tag{44}$$

For large N, evaluate by saddle point. The exponent is $g(q) = \beta N q + \frac{N-3}{2}\ln(1-q^2)$, with $g'(q) = \beta N - (N-3)q/(1-q^2) = 0$, giving

$$q^* = \frac{\beta(1-q^{*2})}{1-3/N} \;\approx\; \beta(1-q^{*2}). \tag{45}$$

For $\beta < 1$: $q^* \in (0,1)$ and the saddle is in the interior. For $\beta = 1$: $q^* = (\sqrt{5}-1)/2 \approx 0.618$ (the golden ratio minus one). The saddle-point exponent $g(q^*)$ determines the leading behaviour of the MGF. When $g(q^*)$ differs from $\beta^2 N/2$ (the Gaussian result), the signal-to-noise threshold α_c shifts. A systematic evaluation of this correction is an important direction for future work.

9 Finite-βN crossover

The results of the preceding sections delineate two regimes:

	Small βN	Large βN
Basins	wide caps, $\theta_c = O(1)$	microscopic, $\theta_c \sim 1/\sqrt{\beta N}$
Geometric overlaps	abundant	exponentially rare
Energy barriers	$O(N/b)$, extensive	$O(1/\beta)$, thermal
Connectivity bottleneck	energy	geometry
Phase transition mechanism	basin percolation	softmax signal-to-noise
α_c	$\frac{1}{2}(1-1/b)^2$	1/2 (S/N), or $\frac{1}{2}\ln(\beta N/8)$ (geom.)

The crossover between the two regimes is determined by the condition

$$\alpha_c^{\text{sphere}}(\beta, N) = \alpha, \qquad \text{i.e.} \qquad \beta N = 8\, e^{2\alpha}. \tag{46}$$

At $\alpha = 0.5$: $\beta N_{\text{cross}} \approx 22$. At $\alpha = 1.0$: $\beta N_{\text{cross}} \approx 59$. At $\alpha = 2.0$: $\beta N_{\text{cross}} \approx 436$.

Finite-N corrections. The leading formula (39) drops polynomial prefactors. Retaining them from (27):

$$\alpha_c^{\text{sphere}} = \frac{1}{2}\ln\frac{\beta N}{8} + \frac{1}{N}\ln\sqrt{\frac{\pi\,\beta N}{4}} + O\Big(\frac{\ln N}{N}\Big). \tag{47}$$

The subleading correction is positive and $O(\ln N/N)$, slightly increasing α_c at finite N.

10 Summary

1. **Geometry first, energy second.** For $b = \beta N$, energy barriers between geometrically overlapping basins are $O(1/\beta)$—thermally irrelevant. The question reduces to: do any caps overlap?

2. **The Gaussian approximation fails catastrophically for $\beta N \gg 1$.** At $q_{\text{geom}} = 1 - 4/(\beta N)$, the Gaussian overestimates the tail probability by $\exp\big[\frac{N}{2}(\ln(\beta N/8) - 1)\big]$, a super-exponential factor.

3. **Geometric percolation requires $\alpha \sim \frac{1}{2}\ln N$.** For any fixed load α, the geometric graph has no edges once $N > 8e^{2\alpha}/\beta$. At $\alpha = 0.5$, $\beta = 1$: no neighbours for $N > 22$.

4. **The transition at $\alpha = 1/2$ is not geometric percolation.** It is driven by signal-to-noise competition in the softmax: $e^{N/2}$ individually negligible patterns collectively overwhelm the target.

5. **Crossover at $\beta N \approx 8\, e^{2\alpha}$.** Below this, the percolation picture from the $b = O(1)$ analysis applies qualitatively. Above it, basins are isolated and the softmax mechanism dominates.